Configuration Management (CM)

CM Overview

Required PA Information

Intent

Manages the integrity of work products using configuration identification, version control, change control, and audits.

Value

Reduces loss of work and increases the ability to deliver the correct version of the solution to the customer.

Additional Required PA Information

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Explanatory PA Information

Practice Summary

Level 1

CM 1.1	Perform version control.

Level 2

CM 2.1	Identify items to be placed under configuration management.
CM 2.2	Develop, keep updated, and use a configuration and change management system.
CM 2.3	Develop or release baselines for internal use or for delivery to the customer.
CM 2.4	Manage changes to the items under configuration management.
CM 2.5	Develop, keep updated, and use records describing items under configuration management.
CM 2.6	Perform configuration audits to maintain the integrity of configuration baselines, changes, and content of the configuration management system.

Additional PA Explanatory Information

Planning for configuration management activities includes controlling work products developed or modified by the project.

The work products placed under configuration management include:

- Deliverables to the customer

- Work products provided by the customer
- Designated internal work products, including:
    - Solution-related components
    - Support components, e.g., procedures
- Acquired solutions
- Systems or tools

Baselines represent an approved version of a work product and provide a clear and accurate understanding for future use. Add new baselines to the configuration management system. Systematically monitor and control changes to baselines and work products using:

- Configuration identification
- Configuration control
- Change management
- Configuration auditing functions of configuration management

External References

External Reference Item	Link
U.S. Department of Commerce, National Institute of Standards and Technology (NIST), Software Security in Supply Chains	https://www.nist.gov/itl/executive-order-14028-improving-nations-cybersecurity/software-security-supply-chains-software-1
U.S. Department of Commerce, National Telecommunications and Information Administration (NTIA), Software Bill of Materials (SBOM)	https://www.ntia.gov/sbom
U.S. Department of Commerce, NTIA, Survey of Existing SBOM Formats and Standards	https://www.ntia.gov/files/ntia/publications/ntia_sbom_formats_and_standards_whitepaper_-_version_20191025.pdf

Related Practice Areas

Verification and Validation (VV)

Context Specific

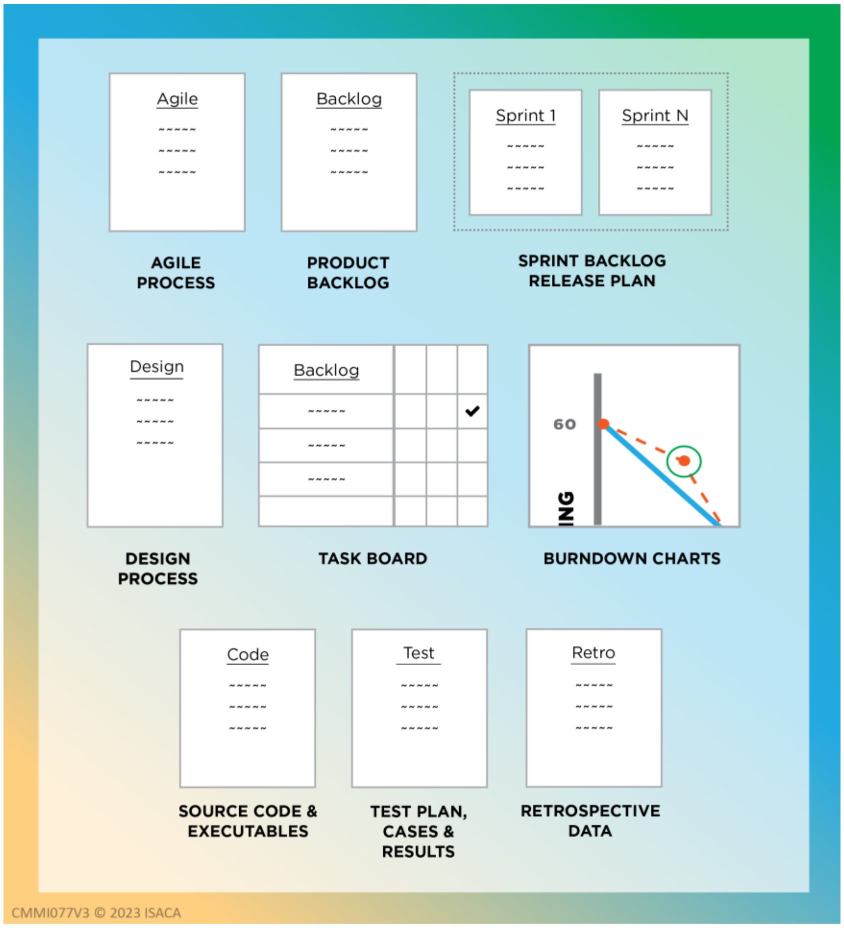
Agile Development

Context Tag:	Agile Development
Context:	Integrate agile techniques and ceremonies with other processes.

Configuration management is performed throughout an agile development project. Figure CM-1: Example Agile Configuration Control Items shows example configuration items for agile development that are placed under selected levels of configuration control. Configuration management processes are used to track the various backlog versions and the ripple effects to design information, code, test cases, and test results.

Configuration management processes are used with agile development efforts to maintain the integrity of work products and deliverables. In an agile development project, the definition of "done" is also a term that the team typically discusses and decides when conducting Sprint planning, reviews, and retrospectives, so that the entire team agrees on the criteria for knowing when the work product or solution is complete. Understanding the definition of "done" is important to being able to verify and validate that the correct versions are produced and delivered from each Sprint.

Figure CM-1: Example Agile Configuration Control Items



Development

Context Tag:	CMMI-DEV
Context:	Use processes to develop quality products and services to meet the needs of customers and end users.

Examples of work products that can be placed under configuration management may include:

- Hardware and equipment
- Drawings, diagrams, and mockups
- Product specifications
- Tool configurations

- Code and libraries
- Compilers
- Test tools and test scripts
- Installation logs
- Product data files
- Product technical publications
- Plans
- User stories
- Iteration backlogs
- Process descriptions
- Requirements
- Architecture documentation and design data
- Product line plans, processes, and core assets

An example of a baseline is an approved description of a product that includes internally consistent versions of requirements, requirement traceability matrices, design, discipline-specific items, and installation, end user, and operations documentation.

For product lines, apply configuration management across the products in the product line and across multiple versions of core assets and products. (Refer to the definition of "product line" in the glossary.)

DevSecOps

Context Tag:	DevSecOps
Context:	DevSecOps is a mindset, a culture, and set of practices that fosters close cooperation between development, operations, and security to plan, develop, test, deploy, release, and keep updated a secure solution.

DevSecOps Configuration Management is highly tool-dependent due to the driving principle to automate routine tasks and the need to integrate development and operations activities to reduce time to market. Many tools and environments used in DevSecOps are not owned, inhouse developed, or on-premises. Configuration management practices can vary dependent on whether the tools and environment are on-premises, cloud, or hybrid, and should be included in their Configuration Management Plan. DevSecOps teams typically use a source repository where the software code is stored. A repository containing code, binaries, and other artifacts is used in conjunction with a configuration management database to track the status of all configuration items. Configuration items may include:

- Requirements documents
- Operational concepts
- Design documents
- Interface descriptions
- Data model descriptions
- System descriptions

- Security Technical Implementation Guide (STIG)
- Workflow diagrams

Configuration audits typically need to be conducted more frequently and iteratively given the nature of the DevSecOps deployment processes and associated changes. Baseline audit reviews should include both the software source code and repository, and also the project documentation, DevSecOps systems, processes, automation steps, and environment as needed.

Level 1

CM 1.1

Required Practice Information

Practice Statement

Perform version control.

Value

Increases customer satisfaction by ensuring that the correct solution is delivered.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Identify the correct versions of work products. This ensures the right versions are available for use or for restoring to a previous version.

Example Activities

Example Activities	Further Explanation
List the work products to be placed under version control and keep it updated.	Include all versions and other relevant information, e.g., location, ownership.
Control versions.	

Example Work Products

Example Work Products	Further Explanation
List of work products and their versions	

Level 2

CM 2.1

Required Practice Information

Practice Statement

Identify items to be placed under configuration management.

Value

Reduces risk of rework and verifies that the right version is delivered to the customer.

Additional Required Information

If Data is included in the selected view: Identify data items, or elements, critical to the delivery of the solution to be placed under configuration management such as personally identifiable information, customer data, operational data, business and financial data, and analytical data.

If People is included in the selected view: Identify people-related work products to be placed under configuration management such as a compensation strategy, workforce competency descriptions, and personnel transition checklists.

If Safety is included in the selected view: Identify hazard and safety-related work products to be placed under configuration management such as hazard analysis, safety improvement plans, safety training materials, and safety related software and hardware.

If Security is included in the selected view: Identify security-related work products to be placed under configuration management such as configurations of network devices, systems, applications, and documentation.

Explanatory Practice Information

Additional Explanatory Information

Based on criteria established during planning, identify configuration items that need to be controlled, managed, and accessed. Logical groupings provide ease of identification and controlled access. Identification typically includes:

- Logical groupings of work products such as:
  - Solutions delivered to the customer
  - Designated internal work products
  - Acquired solutions
  - Tools, equipment, and other assets of the project’s environment
  - Solution documentation and other support materials
- Why and how they are grouped
- How they are controlled, managed, and accessed

Example Activities

Example Activities	Further Explanation
Assign unique identifiers to configuration items.	
Describe the important characteristics for each configuration item.	
Specify when each item is placed under configuration management.	Describe the nature and timing of changes and when and how they may affect the work products or solutions at each stage.

Example Work Products

Example Work Products	Further Explanation
Identified configuration items	Characteristics may include: • Owner or author • Type of work product or solution • Key features • Purpose of the item • Relationships to other items and solutions • Retention • Version • Security

Context Specific

Safety

Context Tag:	CMMI-SAF
Context:	Use processes to incorporate safety considerations as an integral part of the solution, work, project, and organization.

Log, track, and manage safety management work products throughout the solution lifecycle. Develop and keep updated a hazard tracking system with data for each hazard as appropriate for the solution. Examples of applicable data to include in the hazard status log include:

- Type of hazard
- Solution lifecycle phases affected by the hazard
- Causal factor, e.g., hardware, software, and human
- Effects of the hazard
- Initial risk type and associated risk category
- Risk event and associated risk category
- Hazard mitigation measures
- Action person(s) and organizational element
- Hazard status, e.g., open or closed

- Hazard traceability, or running history of actions taken or planned, with rationale to mitigate risks
- Record of risk acceptance(s); risk acceptance authority and user concurrence authority, as applicable, by title and organization, date of acceptance, and location of the signed risk acceptance document(s)
- Reasonably anticipated hazardous materials that are used or generated during the lifecycle of the system, e.g., installation, test and evaluation, normal use, maintenance or repair, and disposal of the system
- Reasonably anticipated hazardous materials to be used or generated in emergency situations, e.g., exhaust, fibers from composite materials released during accidents, and combustion byproducts

Security

Context Tag:	CMMI-SEC
Context:	Use processes to incorporate security considerations as an integral part of the solution, work, project, and organization.

Management of open source components and their security requirements and vulnerabilities is critical since frequently open source components are abandoned or not updated. This can be addressed using a Software or Sales Bill of Materials (SBOM), also known as a Software Bill of Sales (SBOS).

An SBOM enables organizations to effectively track all components in their codebases. Given that open source components are frequently essential components of application development, software development teams can use an effective Software Composition Analysis (SCA) tool to inventory the open source and third-party components in their code.

CM 2.2

Required Practice Information

Practice Statement

Develop, keep updated, and use a configuration and change management system.

Value

Reduces the cost and effort needed to control the integrity of work products and solutions.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Enables consistent and controlled access to work products and solutions and provides the ability to restore a previous version, configuration, or baseline.

A configuration management system:

- May include both manual or automated methods, tools, or complete systems to control work products and solutions
- May be embedded or integrated with other tools to produce work products and solutions
- Includes the procedures for accessing the system

The change management system:

- Provides a means for controlling changes to the identified configuration items
- Includes the storage media, procedures, and tools for recording and accessing change requests
- Is typically integrated with a configuration management system

Example Activities

Example Activities	Further Explanation
Describe how the items and changes to them are controlled, used, and managed throughout the solution lifecycle.	
Establish methods to manage multiple levels of control.	The level of control is typically selected based on work objectives, risk, type, and resources. Example levels of control include: • Uncontrolled: Anyone can make changes • Version controlled: Authors or owners control changes • Baselined: A designated authority authorizes and controls changes, and notifies affected stakeholders
Provide access control to ensure authorized access to the configuration management system.	
Store and retrieve configuration items in the configuration management system.	Typically, storage and retrieval functions in a configuration management system include a check-in and check-out function.
Preserve the contents of the configuration management system.	Examples of preservation functions of the configuration management system may include: • Backup and restoration of configuration management items, e.g., files, database schema, physical artifacts

Example Activities	Further Explanation
	• Archive of configuration management files, consider business, legal, and regulatory requirements for retention and archival • Recovery from configuration management errors • Keeping previous versions according to retention rules
Update the configuration management system as necessary.	

Example Work Products

Example Work Products	Further Explanation
Configuration management system	
Change management system	
Updated configuration items	

Context Specific

DevSecOps

Context Tag:	DevSecOps
Context:	DevSecOps is a mindset, a culture, and set of practices that fosters close cooperation between development, operations, and security to plan, develop, test, deploy, release, and keep updated a secure solution.

DevSecOps teams use their configuration tools to incrementally build an automated Continuous Integration / Continuous Delivery (CI/CD) pipeline for the software that spans initial build through automated releases. A primary benefit of CI is that it saves time during the development cycle by identifying and addressing issues early. It also reduces the time spent on bug fixes and regression by providing continuous feedback to the development team, allowing them to take corrective or preventative action. Automated and built-in configuration audits during CI help uncover issues in a timely manner. The binary code from the various builds is stored in their artifact repository along with supporting artifacts and the configuration management database is used to track the high-level status of all configuration items used.

DevSecOps teams typically use agile project management tools which support agile methodology and CI/CD configuration management, e.g., JIRA, Confluence, Azure Boards, to manage change requests through automated workflows and change states. Tools of this type allow for all information relevant to the change to be captured and used to support decision making, impact analysis, and communicating with stakeholders on the status of their changes as it moves through the change process from request to implementation. Configuration of these tools also needs to be controlled with appropriate change and approval mechanisms. Otherwise, a poor configuration of an automated test tool, analysis tool, build tool, or security scanner may identify false errors or allow defects or security vulnerabilities to go undetected.

CM 2.3

Required Practice Information

Practice Statement

Develop or release baselines for internal use or for delivery to the customer.

Value

Ensures the integrity of the work products.

Additional Required Information

<u>If Data is included in the selected view</u>: Record and keep updated baseline configurations of data management artifacts, e.g., models and representations, data architecture, data management approach, data quality reports.

<u>If Safety is included in the selected view</u>: Record and keep updated baseline configurations and inventories of safety controls for solution components and related documentation throughout the solution lifecycle.

<u>If Security is included in the selected view</u>: Record and keep updated baseline configurations and inventories of security controls for software, hardware, firmware, systems, and related documentation throughout the solution lifecycle.

Explanatory Practice Information

Additional Explanatory Information

Baselines help to formally control changes by establishing points where the status of controlled work products is known and approved. Use changes to develop the next baseline. Typically, a board or a group of people, e.g., the work team, an approval group, or a Change or Configuration Control Board (CCB), formally approves baselines and changes to baselines. The composition of this board may change over time depending on the impacted work products and affected stakeholders. For example, if the baselines or changes are security related, security experts may be invited to the board.

Represent a baseline by assigning an identifier to a configuration item or a collection of configuration items and associated entities at a point in time. As a solution evolves, multiple baselines can exist, and it is important to be able to reproduce any of these multiple baselines.

Example Activities

Example Activities	Further Explanation
Obtain authorization or approval before developing or releasing baselines of configuration items.	An approval group will typically assess the impact and necessity of changes to configuration items.
Develop or release baselines only from configuration items in the configuration management system.	

Example Activities	Further Explanation
Record the set of configuration items contained in a baseline, so that the baseline is reproducible.	
Make the current set of baselines available.	Only stakeholders with approved access should be able to access the baseline information. Additionally, other techniques such as: “build cop,” “don’t break the build,” “release on demand,” “continuous integration,” and “continuous delivery” may be used to manage and control baselines.

Example Work Products

Example Work Products	Further Explanation
Authorization	Includes establishing the baselines and changes to them.
Baselines	Contains the configuration items and the associated changes. Multiple baselines can exist for a single configuration item.

Context Specific

Development

Context Tag:	CMMI-DEV
Context:	Use processes to develop quality products and services to meet the needs of customers and end users.

Include hardware, software, and documentation work products as appropriate in baselines.

One common set of baselines includes the system level requirements, system element level design requirements, and the product definition at the end of development or beginning of production. These baselines are typically referred to respectively as the functional baseline, allocated baseline, and product baseline.

A software baseline can include:

- A unique identifier
- Version release notes
- A set of requirements
- Design
- Interfaces
- Source code files
- Executable code
- Test environments, test cases, etc.
- Build files
- User documentation, such as installation, operational, help, etc.

A hardware baseline can include:

- A unique identifier
- Version release notes
- A set of requirements
- Design
- Interfaces
- Drawings or schematics
- Prototypes, simulations, breadboards, etc.
- Parts and materials
- Test environment
- Build or manufacturing instructions
- User documentation, such as installation, operational, help, etc.

CM 2.4

Required Practice Information

Practice Statement

Manage changes to the items under configuration management.

Value

Reduces costs and schedule impacts by ensuring that only authorized changes are made.

Additional Required Information

If Safety is included in the selected view: Identify, record, and keep updated changes to safety related items placed under configuration management controls, e.g., configurations of network tools, systems, applications, and documentation.

If Security is included in the selected view: Identify, record, and keep updated changes to security configurations of networks, tools, systems, applications, and documentation.

Explanatory Practice Information

Additional Explanatory Information

Analyze change requests to determine their impact on the work product, related work products, budget, and schedule.

Maintain control over the configuration of the work product baseline by:

- Tracking the configuration of each item
- Approving changes to items and baselines
- Approving new configurations and baselines

Example Activities

Example Activities	Further Explanation
Initiate and record change requests.	Typically includes: • Changes to requirements • Failures and defects in work products • Needs from stakeholders, end users, and customers • Description of the impact to work products and solutions
Analyze the impact of change requests.	Analysis should consider impacts to: • Technical and project requirements • Impact beyond the immediate project or contract requirements • Impact on release plans • Cost • Schedule • Quality • Functionality • Commitments
Categorize and prioritize change requests.	Typically includes: • Allowing for emergency changes • Allocating changes to future baselines
Review and get agreement on change requests to be addressed in the next baseline with affected stakeholders.	An approval board typically: • Reviews changes • Records the disposition of each change request and the rationale for each decision • Reports results to stakeholders
Track the status of change requests to closure.	
Incorporate changes in a manner that maintains integrity.	Examples of check-in and check-out include: • Confirming the revisions are authorized • Updating the configuration items • Maintaining versions of work products • Archiving the replaced baseline and retrieving the new baseline • Commenting on the changes made • Assigning changes to related work products
Perform reviews or testing to ensure changes have not caused unintended impacts.	Examples: • Automated unit test • Regression testing • Performance testing
Record changes to configuration items and rationale.	

Example Work Products

Example Work Products	Further Explanation
Change requests	Typically includes: • Description of the change • Category of change • Priority of change • Status of change • Impact of change • Estimated implementation time • Actual implementation time
Results of change impact analysis	
Approval board records	
Revision history of configuration items	
Results of reviews or tests for unintended impacts	
Revised work products and baselines	

CM 2.5

Required Practice Information

Practice Statement

Develop, keep updated, and use records describing items under configuration management.

Value

Reduces rework through accurate descriptions of the configuration items and status of changes.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Record the configuration actions and status to enable recovery to previous versions and to understand the status of the item and the changes that were or are being made.

Example Activities

Example Activities	Further Explanation
Record configuration management actions in sufficient detail so the content and status of each	

Example Activities	Further Explanation
configuration item is known and previous versions can be recovered.	
Ensure that affected stakeholders have access to and knowledge of the configuration status of configuration items.	Examples of activities for communicating configuration status include: • Providing access permissions to authorized users • Making baseline copies readily available to authorized users • Automatically alerting affected stakeholders when items are checked-in, checked-out, or changed, and of decisions made regarding change requests
Specify the differences among previous, related, and latest versions of baselines.	These are often referred to as release notes.
Identify the version of configuration items that constitute a specific baseline.	Also, identify the changes used to develop that baseline.
Revise the status and history, e.g., changes, of each configuration item as necessary.	

Example Work Products

Example Work Products	Further Explanation
Revision history or change log of configuration items	
Change request records	
Status of configuration items	
Differences between baselines	

Context Specific

Data

Context Tag:	CMMI-DATA
Context:	Use processes to incorporate data management best practices as an integral part of the solution.

Record, manage, and keep updated key information for data items, e.g., authoritative source, ownership and governance data parameters, metadata, access and distribution parameters, privacy classification, security classification, retention parameters.

CM 2.6

Required Practice Information

Practice Statement

Perform configuration audits to maintain the integrity of configuration baselines, changes, and content of the configuration management system.

Value

Increases customer satisfaction and stakeholder acceptance by ensuring that the customer receives the agreed-on and correct versions of work products and solutions.

Additional Required Information

<u>If Safety is included in the selected view</u>: During configuration audits, include verification of integrity and correctness of safety aspects of the solution components.

<u>If Security is included in the selected view</u>: During configuration audits, include verification of integrity and correctness of security aspects of the solution components.

Explanatory Practice Information

Additional Explanatory Information

Configuration audits confirm:

- Configuration management records and configuration items are complete, consistent, and accurate
- Integrity of the baselines, change requests, and associated items

Example Activities

Example Activities	Further Explanation
Assess the integrity of baselines and generate action items to address identified issues.	Examples include: • Physical work product reviews verifying changes • Functional work product reviews verifying changes • Comparison of approved changes versus actual changes made in a work product
Confirm integrity of configuration management records.	Typically, this includes confirming: • Correctly identified configuration items • The completeness, correctness, and consistency of items
Review the structure and integrity of items in the configuration management system.	
Record action items and track them to closure.	

Example Work Products

Example Work Products	Further Explanation
Configuration audit or review results	Contains the objectives for and outcomes of audits in enough detail to take action.
Action items	Describes actions needed to address findings from audits and the criteria needed to determine when they can be closed.